

AI_TOOLS_AND_USAGE

AI Tools & Usage Documentation

1. AI/ML Stack

Component	Library / Provider	Model	Purpose
LLM	Groq API (<code>groq</code> Python SDK)	<code>openai/gpt-oss-20b</code>	Tutoring, concept extraction, quiz generation/grading, recommendations
Embeddings	<code>sentence-transformers</code>	<code>all-MiniLM-L6-v2</code>	Turns material chunks and tutor questions into vectors
Vector store	ChromaDB	—	Stores/retrieves embedded chunks per project for the Tutor
PDF text extraction	PyPDF2	—	Extracts per-page text from uploaded PDFs before chunking/embedding

All LLM calls go through one function, `llm_service.call_llm`. Every call uses the same fixed setup: model `openai/gpt-oss-20b`, a single `user`-role message (no system prompt), `max_tokens=1024`, and JSON-mode output only where the feature needs structured data. No `temperature`, `top_p`, or `stream` overrides are used.

2. Where AI Is Used

Concept Extraction — After a material is chunked and embedded, its first few chunks are sent to the LLM to identify 4–6 main concepts, which are stored and each seeded with a starting mastery score of 30%.

Tutor (RAG) — A student's question is embedded and matched against the project's material chunks in ChromaDB. If no sufficiently relevant chunk is found, the app replies with a fixed "not enough information" message and skips the LLM entirely. Otherwise, the matched chunks (with material name + page number) are passed to the LLM as context, along with a short note on the student's weak concepts, and the answer is returned with its sources cited.

Quiz Generation — A concept is picked with a mastery-weighted random selection (weaker concepts come up more often), and the LLM generates one multiple-choice or open-ended question calibrated to the student's mastery level for that concept.

Quiz Grading — Multiple-choice answers are graded by plain string comparison, no LLM involved. Open-ended answers are graded by the LLM, which returns a correct/incorrect verdict, short feedback, and any

missing concepts.

Growth Recommendations — Once a student has mastery history, the LLM turns their concept trends into 1–2 sentences of specific, encouraging guidance. This runs automatically when a quiz is completed, and on demand from the Analytics tab.

3. AI Observability

Every LLM call is logged with its feature, model, latency, token counts, and success/failure status. This log powers per-project Analytics, the Admin "AI Usage" dashboard (requests, tokens, latency, estimated cost), and Admin "System Health" (recent error rate, recent failures).

4. Heuristic AI-Quality Evaluation

An Admin "Evaluation" view computes quality signals purely from logged data (no human-labeled eval set): Tutor groundedness rate and unsupported-question handling rate, quiz generation/grading failure rates, and recommendation failure rate. Snapshots can be saved for comparison over time.

5. Vector Storage

Each project has its own ChromaDB collection, so retrieval never crosses project boundaries. Chunks are embedded with `all-MiniLM-L6-v2`, with material name and page number stored alongside each embedding — this is what lets the Tutor cite its sources.